G H Raisoni International Skill Tech University, Pune

(Established by Maharashtra Act No. IX of 2024 and under section 2(f) of the UGC Act 1956)

FY B. Tech (Summer /SEM II)

ESE SUMMER 2026 (2025 Pattern)

Digital Logic Design (25UESL1105)

[Time: 2.30 Hours] **[Max. Marks 60]**

Instructions to the candidates:

1) ***All questions compulsory.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q.No.	*Sub Question*		*Marks*	*CO*	*BL*
		For Q-1, Q-2 and Q-3—All sub questions are compulsory.			
Q. 1	*a)*	Convert the decimal number $(50)_{10}$ into following equivalent numbering system i) Binary ii) Excess-3 code iii) Hexadecimal iv) BCD	**[4]**	**CO1**	**L3**
	b)	Explain canonical SOP and canonical POS forms. Illustrate each with suitable examples.	**[4]**	**CO2**	**L2**
	c)	Differentiate between Mealy and Moore machine also write its features and their applications.	**[4]**	**CO3**	**L2**
Q. 2	*a)*	Perform the following subtraction using 2's complement method: $(85)_{10} - (28)_{10}$	**[4]**	**CO1**	**L3**
	b)	Differentiate between combinational circuit and sequential circuit with suitable example.	**[4]**	**CO2**	**L2**
	c)	Simplify the Boolean function using K-map $F(A, B, C, D) = \sum m(0,2,4,5,6,7,13,15)$	**[4]**	**CO2**	**L3**

Q. 3	a)	Explain the operation of a Full adder combinational circuit, support your explanation with the truth table, Boolean expression for sum & carry & logic diagram.	[4]	CO2	L2
	b)	Using Boolean, Prove that (A+B)(A+C) = A+BC , write all steps.	[4]	CO1	L3
	c)	Convert the given binary number $(11001011)_{)2}$ into following equivalent numbering system. i) Decimal ii) Gray code iii) Octal iv) BCD	[4]	CO1	L3

For Q-4 and Q-5, Sub question a) is compulsory. And Solve any one between b) and c)

Q. 4	a)	Explain working with neat diagram of JK flip flop also write its applications.	[6]	CO3	L2
	b)	Explain following with respect to Algorithmic State Machines (ASM) i) State diagram ii) State table iii) State reduction & assignment	[6]	CO3	L2
		OR			
	c)	Highlight any five applications of Finite State Machines (FSM) with brief explanations.	[6]	CO3	L2
Q. 5	a)	Explain shift registers in sequential circuit? Explain their working principle and discuss the different types of shift registers with brief description.	[6]	CO3	L2
	b)	Describe concept of Entity and Architecture used in VHDL.	[6]	CO4	L2
		OR			
	c)	Explain the following elements of VHDL: i) Literals ii) Data types iii) Logic systems	[6]	CO4	L2

$BD + \bar{A}B\bar{C} + \bar{A}BC + \bar{A}\bar{B}\bar{D}$